POWER STUDY 0.1 – 5V / 0.8A SYSTEM TARGET, 2A CONVERTER RATING

Proposal only: not installed or routed on PCB B v0.4. Read README.md before integration.
Protected 24V input. Common ground proposal replaces the external CC/CV module and its possible shunt.

J1
FROM B / protected input
VIN_PROTECTED 1
GND 2

C1
4.7uF / 100V X7S
VIN_PROTECTED
GND

C2
4.7uF / 100V X7S
VIN_PROTECTED
GND

C3
220nF / 100V X7R
VIN_PROTECTED
GND

#FLG1
PWR_FLAG
VIN_PROTECTED

#FLG2
PWR_FLAG
GND

#FLG3
PWR_FLAG
+5V_BUCK

J2
TO B / F2 INPUT
+5V_BUCK 1
GND 2

L1
10uH / XAL6060-103MEC
SW
+5V_BUCK

R1
100k / 0805
VIN_PROTECTED
BUCK_EN

U1
LMR36520ADDAR
VIN_PROTECTED 2 → VIN
BUCK_EN 3 → EN
VCC_INTERNAL 6 → VCC
GND 1 → GND
BOOT 7 ← BOOT
SW 8 ← SW
FB 5 ← FB
PG 4 ← PG_5V_TP

C4
100nF / 25V
BOOT
SW

C6
22uF / 25V X5R
+5V_BUCK
GND

C7
22uF / 25V X5R
+5V_BUCK
GND

C8
22uF / 25V X5R
+5V_BUCK
GND

C9
22uF / 25V X5R
+5V_BUCK
GND

JP1
CLOSE TO DISABLE BUCK
BUCK_EN 1
GND 2

C5
1uF / 25V
VCC_INTERNAL
GND

R2
100k / 1%
+5V_BUCK
FB

R3
24.9k / 1%
FB
GND

R4
100k / 1%
+5V_BUCK
PG_5V_TP

OUTPUT: +5V_BUCK -> existing F2 (1A) -> +5V_AB -> A/B electronics. Heater power stays on protected 24V.
VCC_INTERNAL is not an external 5V output. PG_5V_TP is a 5V-domain scope point, not ESP32-safe.
Final integration must pull PG to 3V3 if used by MCU, and deliberately connect logic/power returns; no CC shunt remains.

Nominal output: 1V * (1 + 100k / 24.9k) = 5.016V. 4 x 22uF output bank; simulate 44uF effective worst case.
Model: TI SNVMBX7A / LMR36520A_TRANS, unencrypted PSpice. Temperature and enclosure cooling are not modeled.
Native ngspice attempts are documented as NON-CONVERGENT, not successful functional validation.

No manufacturing release: PCB layout, capacitor DC bias, thermal pad/vias, USB power sharing, sourcing and bench tests pending.

Sheet: /
File: buck_5v.kicad_sch
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